Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/28/2007
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNC06AA32A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2006	**End Date:**	09/30/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNC06AA32A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effort is a collaborative effort between the three Co-PI's and staff at NASA Glenn (Dr. David Urban and Mr. Dennis Stocker). The division of labor is roughly broken up with PBS performing microgravity experiments at the NASA Glen Research Center and numerical simulations at University of Maryland, BHC performing analytical work at University of Hawaii and RLA performing normal gravity experiments at Washington University. The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial		

	performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed 'Flame Design' as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The following is a summary of the research activities during this period. 1) Work has continued on radiative extinction of spherical diffusion flames, which are being investigated experimentally and numerically. The experiments involved microgravity spherical diffusion flames burning ethylene and propane at 0.98 bar. Both normal (fuel flowing into oxidizer) and inverse (oxidizer flowing into fuel) flames are studied, with nitrogen supplied to either the fuel or the oxygen. Flame conditions were chosen to ensure that the flames extinguished within the 2.2 s of available test time; thus extinction occurred during unsteady flame conditions. Diagnostics included color video and thin-filament pyrometry. The computations, which simulated flow from a porous sphere into a quiescent environment, include detailed transport properties, detailed chemistry and an optically thick radiative heat loss model to study the transient radiative extinction of spherical diffusion flames supported by a porous spherical burner. A flow of a reactant (fuel or oxidizer) issued from a porous sphere into a quiescent environment of the other reactant was considered in the simulation. Radiative extinction was observed experimentally and simulated numerically. Extinction time, peak temperature, and radiative loss fraction (ratio of the radiative heat loss rate to the heat generation rate) were found to be independent of flow rate except at very low flow rates. Radiative heat loss was dominated by the combustion products downstream of the flame and was found to scale with flame surface area, not volume. For large transient flames the heat release rate also scaled with surface area and thus the radiative loss fraction was largely independent of flow rate. Peak temperatures at extinction onset were about 1100 K, which is significantly lower than for kinetic extinction. One observation is that while radiative heat losses can drive transient extinction, this is not because radiative losses are increasing with time (flame size) but rather because the heat release rate is falling off as the temperature drops. 2) A new numerical study of limiting conditions for soot particle inception in spherical diffusion flames has been initiated. The flames were modeled using a one-dimensional, time accurate diffusion flame code with detailed chemistry and transport and an optically thick radiation model. Seventeen normal and inverse flames were considered, covering a wide range of stoichiometric mixture fraction, adiabatic flame temperature, and residence time. These flames were previously observed to reach their sooting limits after 2 s of microgravity. Sooting-limit diffusion flames with residence times longer than 200 ms were found to have temperatures near 1190 K where C/O = 0.6, whereas flames with shorter residence times required increased temperatures. Acetylene was found to be a reasonable surrogate for soot precursor species in these flames, having peak mole fractions of about 0.01. 3) Our previous studies have shown a linear relationship between the limit flame temperature and stoichiometric mixture fraction for coflow flames at their sooting limit. The sooting limit in coflow flames is obtained by varying the amount of inert gas until soot appears above a predefined height. In this work sooting limits based on particle luminosity were measured as a function of Zst. For each limit flame, the flame temperature was measured and defined as the limit flame temperature. The limit flame temperature increases linearly with Zst, for both normal and inverse flames. The effects of residence time have also been investigated, and the soot inception temperature was found to be dependent on fuel stream velocity for both the normal and inverse configuration. Earlier experimental results and modeling indicated that a critical carbon to oxygen (C/O) ratio exists in the fuel rich region of the flame where soot inception is no longer favorable due to the presence of oxygen containing species. We extended this model to include the effects of finite rate chemistry and thus residence time on the sooting limit. This involved considering two boundaries for the soot inception zone, one dictated by the C/O ratio and the other by the finite rate chemistry of fuel pyrolysis and soot inception. By combining these two boundaries, a simple model was developed that describes how a sooting limit is attained. By utilizing measured limit temperature data with the model we found the limit inception temperature for soot formation to be 1640 K and the critical carbon to oxygen ratio to be 0.53. 4) Performed analytical analysis of the kinetic and radiative extinction limits of a steady spherical diffusion flame supported by a porous burner. A single step, overall and irreversible reaction that follows high activation energy Arrhenius kinetics was adopted to describe the reaction rate. An optically thin radiation model was used to describe the radiative heat loss. The kinetic and radiative extinction limits were observed from the analysis. Results show that the flame is easier to extinguish when the radiation intensity is higher, as expected, and there exists a maximum radiation intensity above which steady burning is not possible.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Santa KJ, Sun Z, Chao BH, Sunderland PB, Axelbaum RL, Urban DL, Stocker DP. "Numerical and experimental observations of spherical diffusion flames." Combustion Theory and Modelling 2007 Aug;11(4):639-52. http://dx.doi.org/10.1080/13647830601161567 , Aug-2007
Articles in Peer-reviewed Journals	Kumfer BM, Skeen SA, Chen R, Axelbaum RL. "Measurement and analysis of soot inception limits of oxygen-enriched coflow flames." Combustion and Flame 2006 Nov;147(3):233-42. http://dx.doi.org/10.1016/j.combustflame.2006.08.004 , Nov-2006
Awards	Sunderland PB, Urban DL, Stocker DP, Chao BH, Axelbaum RL. "Third Prize, Art Competition Poster for, “Spherical Ethylene Diffusion Flame in Microgravity,“ 5th U.S. Combustion Meeting, San Diego, March, 2007." Mar-2007
Papers from Meeting Proceedings	Santa KJ, Sun Z, Chao BH, Sunderland PB, Axelbaum RL, Urban DL, Stocker DP. "Numerical and experimental observations of spherical diffusion flames." 45th Aerospace Sciences Meeting, Reno, Nevada, January 8–11, 2007. 45th Aerospace Sciences Meeting, Paper No. AIAA-2007-736, 2007. , Jan-2007

Papers from Meeting Proceedings	Santa KJ, Chao BH, Sunderland PB, Urban DL, Stocker DP, Axelbaum RL. "Radiative extinction of gaseous spherical diffusion flames in microgravity." 5th U.S. Combustion Meeting, San Diego, California, March 25–28, 2007. 5th U.S. Combustion Meeting, Paper No. A06, 2007. , Mar-2007
Papers from Meeting Proceedings	Wang Q, Chao BH. "Kinetic and radiative extinctions of sperical diffusion flames." 5th U.S. Combustion Meeting, San Diego, California, March 25–28, 2007. 5th U.S. Combustion Meeting, Paper No. A08, 2007. , Mar-2007
Papers from Meeting Proceedings	Lecoustre VR, Skeen SA, Chao BH, Sunderland PB, Urban DL, Stocker DP, Axelbaum RL. "A computational and analytical investigation of sooting limitsof sperical diffusion flames." 5th U.S. Combustion Meeting, San Diego, California, March 25–28, 2007. 5th U.S. Combustion Meeting, Paper No. F23, 2007. , Mar-2007